Part:	**1.4 – COMPOSITE MATERIALS**
Chapter:	**1422**
Title:	**RESIDUAL STRENGTH AFTER EDGE IMPACT IN COMPOSITES**
Revision:	**0.2**
Date:	**23/01/2024**

1 INTRODUCTION

1.1 OBJECT

The aim of this chapter is to define and present a closed form approximation of the residual strength problem in laminated plates after an edge impact. The low velocity edge impact is that caused by a blunt object with a trajectory normal to the plate edge and in the plane of the plate. The BVID threshold of visibility for this case is conveyed, within Airbus, to be defined for the Detailed Visual Inspection (DVI) as that resulting in a dent depth of 0.6mm after impact (or 0.3mm after relaxation) or an edge crack of 30mm (see Ref. 3 or Ref. 4).

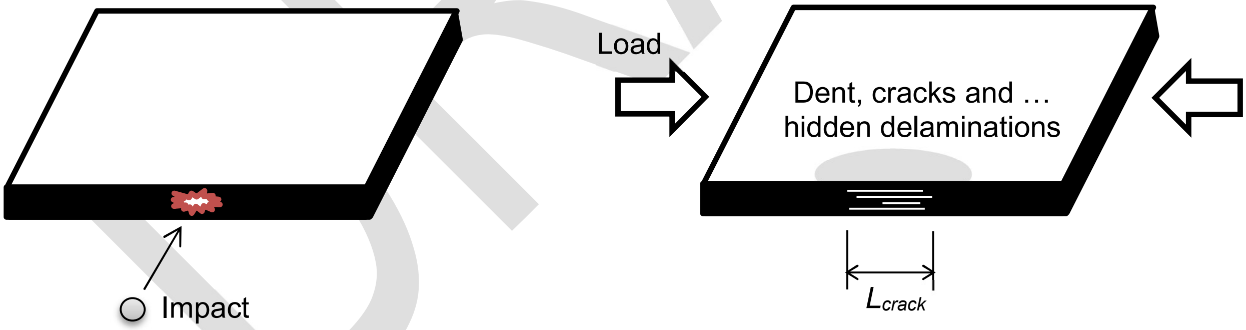

Figure 1. Sketch of the CAIE problem (straight edges)

1.2 SCOPE AND LIMITATIONS

Method applicable to:

- ✓ Fibrous composite solid laminates (made of UD-tape or biaxial-fabrics)
- ✓ Stacking sequences following the typical design rules

- ✓ Low velocity impacts causing BVID damages
- ✓ Straight edges of plates/shells

Limitations:

- ✓ Impacts at/near curves edges, and transverse impact at interior areas of plates/shells, are not covered by the model presented in this chapter

2 METHOD DESCRIPTION

2.1 Generalities

See discussion in chapter 1421 “RESIDUAL STRENGTH AFTER TRANSVERSE IMPACT IN COMPOSITES”.

As mentioned in the introductory section, in addition to the dent depth, the length of the edge crack caused by the impact will be also a valid limit to set the BVID threshold: an edge crack of minimum 30mm can be detected during a DVI. This damage characteristic shall be frequently the criterion to fix the BVID threshold.

2.2 Tension After Edge Impact (TAIE)

As for the case of the transverse impact, the *TAIE* residual strength, in terms of strain, will be considered a material constant for any laminate.

TAIE allocable is expected in general to be greater than the *TAI* design value (after penetrating transverse impact) as defined in chapter 1421. Using conservatively *TAI* as common design value shall be a convenient simplification in most cases.

2.3 Compression After Edge Impact (CAIE)

The proposed model consists of a product of constant parameters and univariate (uncoupled) functions.

$$CAIE = K_B \cdot EKDF \cdot CAIE_{ref} \cdot min[1 + A \cdot (t - t_{ref}), B] \cdot (\, 3 / K_{TOH}\,)^C \qquad (1)$$

…where:

- ✓ K_B is the B-basis correction factor
- ✓ *EKDF* is the applicable Environmental Knock Down Factor
- ✓ $CAIE_{ref}$ is the *CAIE* residual strength after edge impact, at the BVID level, of the QI laminate of 4-5mm thickness, in με [2]
- ✓ K_{TOH} is the stress concentration factor in the laminate due to an open circular hole
- ✓ t is the thickness of the laminate, in mm
- ✓ *A, B*, and *C* are parameters of the model to be adjusted from tests data

Notes:

(1) This methodology is inspired in Airbus Commercial method (Ref. 1 and Ref. 2), but adapted to the preferred reference values (the QI laminate of 4-5mm thickness typically tested during qualification) and the form functions for accounting effect of laminate directionality used in chapter 1421.

(2) The reference allowable $CAIE_{ref}$ strain should cover not only edge impacts but also transverse impacts at or near the edge (with the same definition of BVID threshold)

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	*Units*	*Definition*
BVID	-	Barely Visible Impact Damage, either determined by the visibility threshold, or by the impact energy cut-off. The detectability threshold for edge impacts is conveyed (within Airbus) to be set for the Detail Visual Inspection (DVI), as 1.0mm dent depth after impact (0.3mm after relaxation), or 30mm long edge crack. On the other hand, the energy cut-off is typically set in 35 J, in general function of the part location in the airframe.
CAIE	MPa , με	Residual Compression strength After Edge Impact in a fibrous composite laminate including a BVID
CFRP	-	Carbon Fibre Reinforced Polymer/Plastic
DET/DVI	-	Detailed Visual Inspection
E	MPa , J	Elastic (Young's) Modulus or impact energy, from context
EKDF	-	Environmental knock down factor, i.e. the correction factor to apply to a property obtained at room temperature in order to get the value at applicable hot/wet conditions
K_B	-	B-basis knock down factor (ratio B-basis to mean)
K_{TOH}	-	Stress concentration factor of a circular open hole in an infinite composite plate subjected to axial in-plane load
QI	-	Quasi-Isotropic laminate (typically with 25% plies at 0º, 25% at 90º, 25% at 45º, and 25% at -45º)
RT	-	Room Temperature ambient conditions
t	mm	Thickness of the laminate
TAIE	MPa , με	Residual Tension strength After Edge Impact in a fibrous composite laminate including a BVID

REFERENCES

	Document Reference	Issue	Title
Ref. 1	[Airbus] V57RP1627379	Is. 1.0	A350XWB-2000 Stressing and Design Criteria for LAGRANGE MDO
Ref. 2	[Airbus] V57RP1039013	Is. 4.0	Edge Impact Damage Tolerance Method for A350
Ref. 3	[Airbus] ME0905804	Is. 1.0	PRELIMINARY UPDATE TO BVID REQUIREMENTS FOR EDGE IMPACTS ON A350 XWB COMPOSITE STRUCTURES
Ref. 4	[Airbus] AITM1-0076	Is. 1	Airbus Test Method Fibre. Reinforced Plastics. Determination of compression and tension strength after edge impact

SOFTWARE

Program	Version	Description

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub-chapters affected
0.2 23/01/2024	G. Baños	First issue	All pages

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Structural Analysis	-	Florian Glock ??/01/2024
Stress Methods & Optimization	G. Baños 23/01/2024	-

Table 3. *Approval signatures table for last revision of this chapter*